STATISTICAL AND MACHINE LEARNING-BASED DETECTION OF DOS/DDOS ATTACKS IN IOT NETWORK TRAFFIC

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Link of the project:
https://github.com/andrealabelli/Network_Security_Project/tree/main

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Introduction

- Networked systems have become a fundamental component of modern society
- Increasing number of connected devices
- IoT devices have significantly increased the scale and complexity of modern network environments
- Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks represent a major concern

Dataset used

- “TON_IoT_Train_Test_Network” Dataset
- Kaggle is the world’s largest platform for data science and machine learning
- Used by researchers and practitioners as a benchmark source for developing and evaluating data-driven security solutions
- Multiple attack categories with multiple features such protocols, connection states, packet counts and byte counts present in dataset

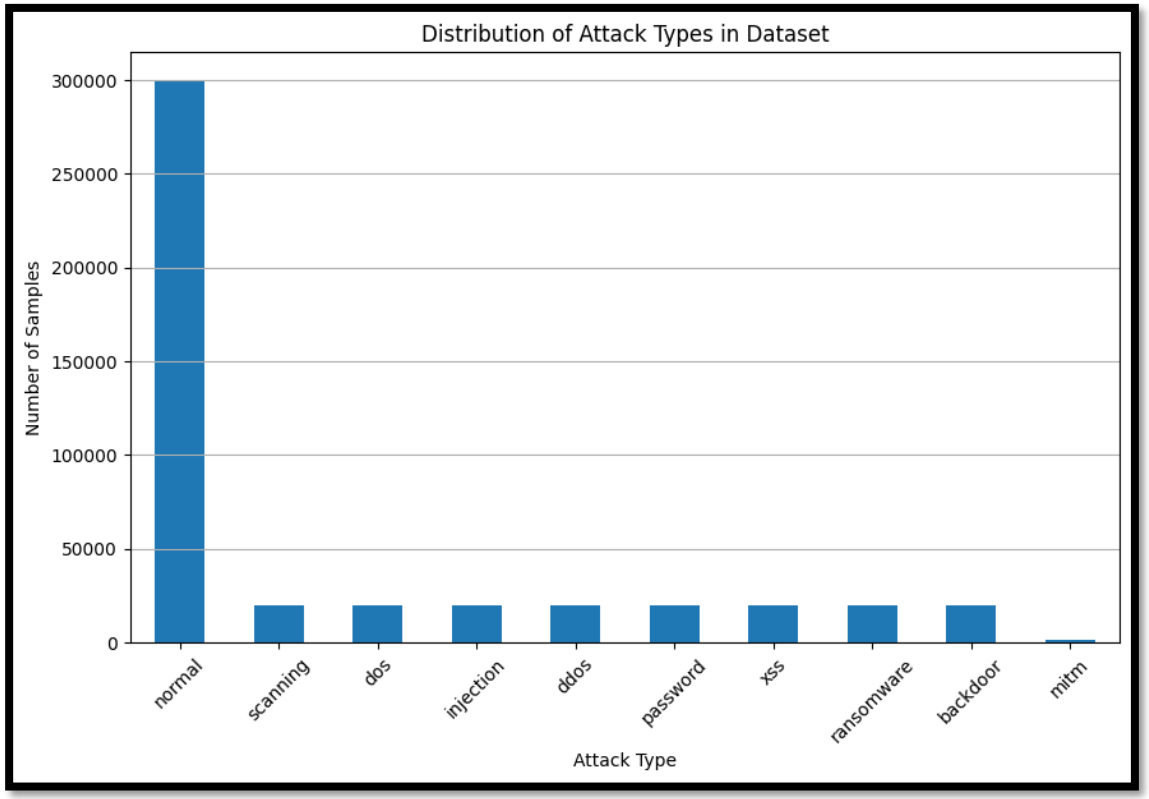
Project Objective

- Analyze statistical characteristics of DoS/DDoS traffic vs “Other” traffic categories in the “TON_IoT” dataset
- TCP traffic and UDP traffic evaluation
- Attack-type comparative analysis
- Three supervised ML algorithms used: Random Forest, Logistic Regression and SVM
- Evaluate impact of original imbalanced dataset and application of SMOTE on the models

Preprocessing

Data preparation

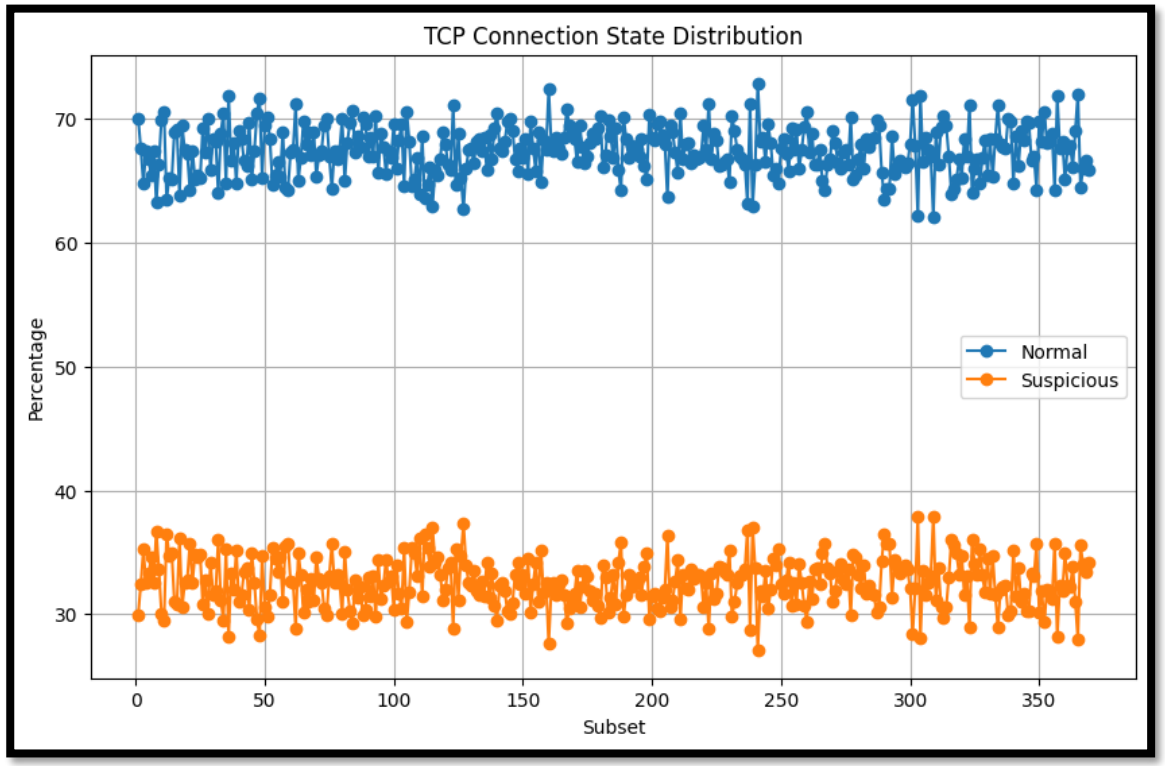
- Distribution of attack types in dataset
- Dataset shuffling to ensure unbiased sampling
- Train/test split (80/20)
- Subset analysis of 1000 samples each



TCP Flooding Analysis

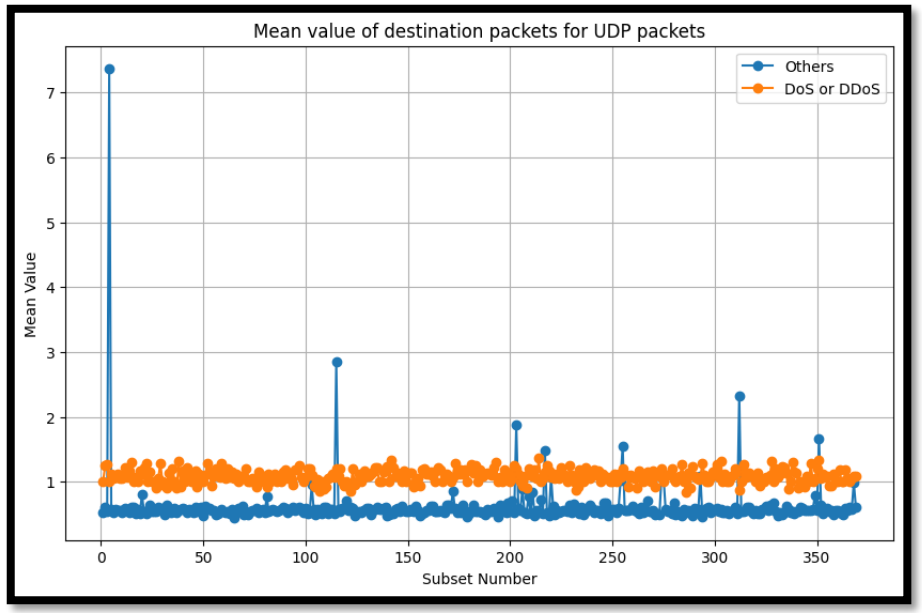
- Connection state mapping: Suspicious vs Normal
- 30-35% Suspicious connections
- 70-75% Normal connections
- Derived TCP threshold at 50%
- TCP flooding attacks alter the distribution of connection states

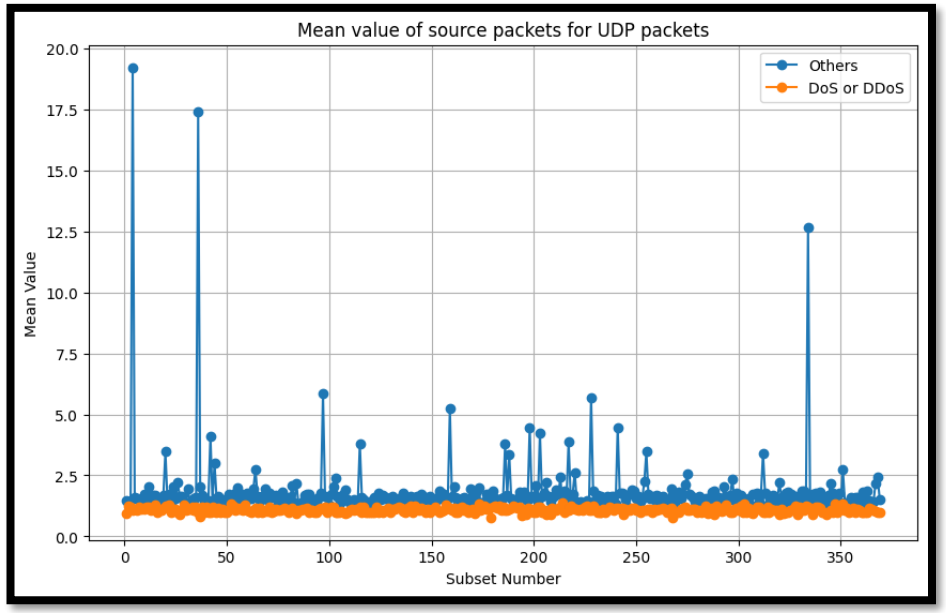
TCP State	Category
S0	Suspicious
REJ	Suspicious
RSTO	Suspicious
RSTR	Suspicious
SH	Suspicious
SF	Normal
S1	Normal
S2	Normal
S3	Normal
OTH	Normal



UDP Flooding Analysis - Packets

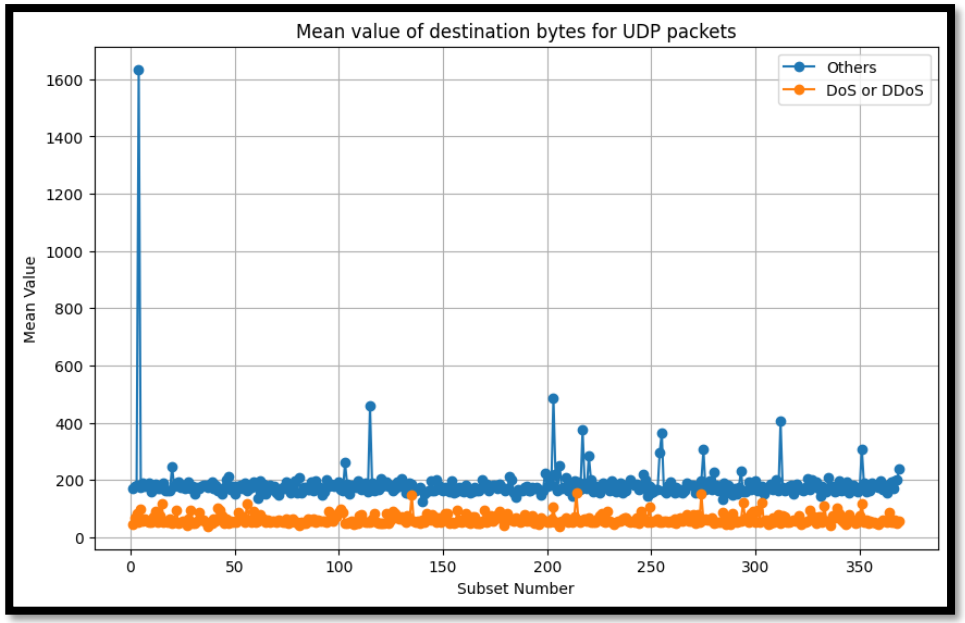
- Destination packet threshold of 0.86
- Source packet threshold of 1.48
- 47% of destination packets exceeds threshold
- 78.7% of source packets fall below threshold

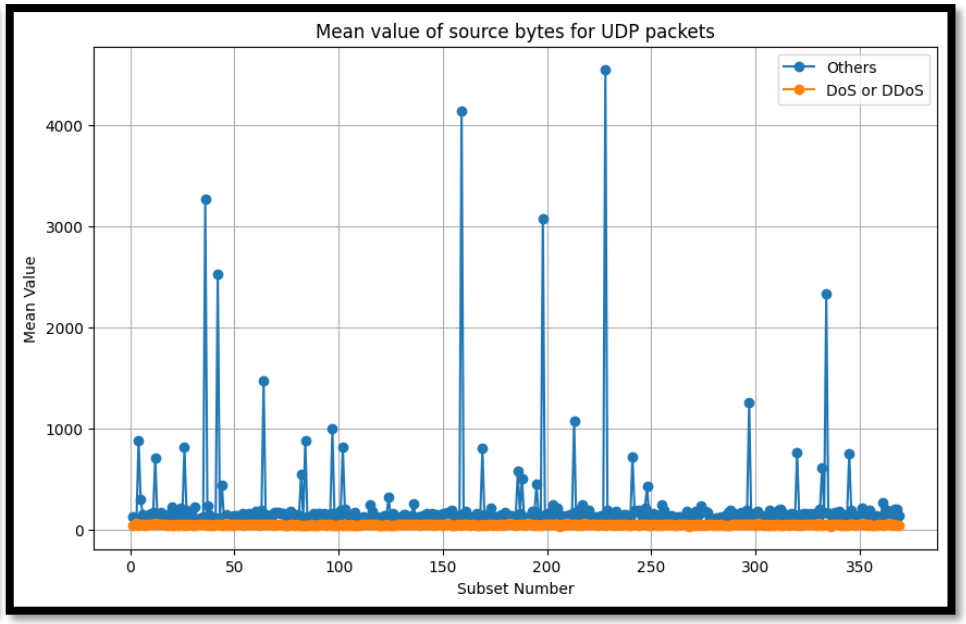




UDP Flooding Analysis - Bytes

- Destination bytes threshold of 124.39
- Source bytes threshold of 139.81
- 67.6% of destination bytes fall below threshold
- 91.5% of source bytes fall below threshold





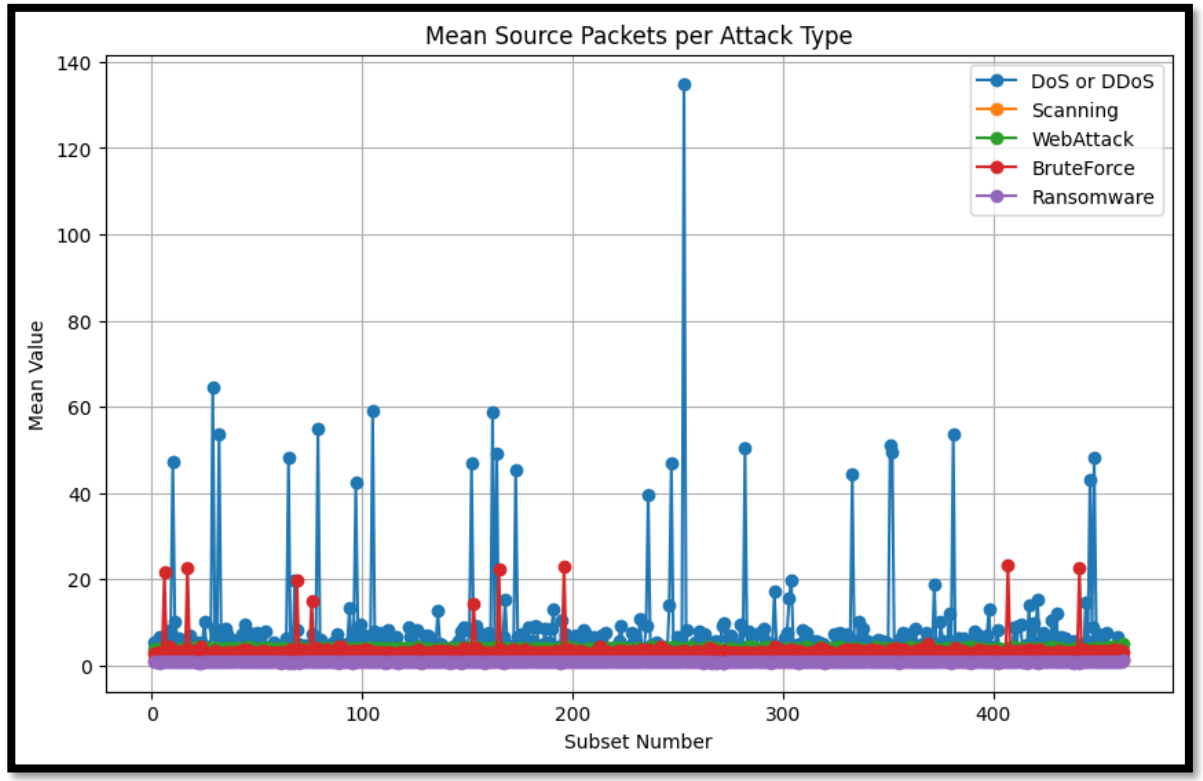
Attack-Type Comparative Analysis – Source Packets

Volumetric Comparison Across Attacks

- DoS/DDoS
- Scanning
- Brute Force
- WebAttack
- Ransomware

- DoS/DDoS traffic exhibits mean source packet values reaching peaks above 100 packets per subset in extreme cases

- “Brute Force and “WebAttack” traffic generally remain below 10–20 packets per subset



Machine Learning Models

- Random Forest is an ensemble learning method based on the combination of multiple decision trees
- Logistic Regression provides a simple yet effective linear baseline for binary classification problems
- Support Vector Machine aims to find an optimal decision boundary that maximizes the margin between different classes

Metrics used

- Accuracy
- Recall
- F1-Score

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Original dataset results

- Random Forest achieves the highest overall accuracy (95.82%) and significantly outperforms the other models in terms of recall and F1-score.

- Logistic Regression and SVM reach high accuracy values, but extremely low recall and F1-score demonstrate that always misclassify samples

- Direct consequence of the imbalanced nature of the dataset and difficulty in modeling non-linear patterns

Model	Accuracy	Recall	F1-score
Random Forest	0.9582	0.5547	0.6935
SVM	0.9148	0.0022	0.0044
Logistic Regression	0.9148	0.0021	0.0041

SMOTE balancing dataset results

- Oversampling technique that generates synthetic samples for the minority class

- Given a minority class sample *xi* and one of its k−nearest neighbors *xj* , a new synthetic sample *xnew* is generated, where λ ∈ [0, 1]

- Recall improved for all the models, but linear models still limited and SVM continues to underperform

$$x_{\text{new}} = x_i + \lambda \cdot (x_j - x_i)$$

Model	Accuracy	Recall	F1-score
Random Forest	0.9192	0.9795	0.6740
SVM	0.1228	0.2623	0.0486
Logistic Regression	0.3856	0.7550	0.1734

Conclusions and future work

- Combining statistical traffic analysis with ensemble machine learning models represents an effective strategy for DoS/DDoS detection

- Applying SMOTE, all models showed significant improvement in recall

- Real-time deployment and evaluation in operational environments represent important directions for future research

Thanks for the attention!